Quantifying the Influence of Socioeconomic and Public Health Predictors on Life Expectancy

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Abstract

This study quantifies the independent influence of socioeconomic and public health predictors on Life Expectancy across 183 countries in the year 2015. Utilizing Multiple Linear Regression (MLR), the analysis addressed multicollinearity and employed backward stepwise regression to develop a statistically robust and parsimonious predictive model. The final model achieved an exceptionally high Adjusted R^2 of $\mathbf{0.8715}$, indicating that over 87% of the variability in Life Expectancy is collectively explained by the seven retained, significant predictors: Adult Mortality Rate, Average Years of Schooling, HIV Death Rate, Early Childhood Thinness, Polio Immunization Deficit, GDP per capita, and Economic Status.

The results highlight that health burdens, specifically the Adult Mortality Rate and the HIV Death Rate, exert the largest and most immediate negative pressures on longevity. Conversely, Average Years of Schooling emerged as the most important positive socioeconomic factor, with each additional year of schooling correlating with a 0.701 years increase in life expectancy, all else being constant. The findings confirm that enhancing global longevity requires coordinated, evidence-based policy interventions focused on strengthening public health infrastructure, expanding educational attainment, and mitigating the core burdens of disease and malnutrition.

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Chapter 1

Introduction

1.1 Background and Motivation

Life Expectancy is globally recognized as the paramount measure of a country's overall development, reflecting not just the health of its citizens but also the efficacy of its social, economic, and political systems. Global efforts to increase longevity require a clear understanding of the specific factors driving success and failure. While it is widely accepted that factors such as national wealth, educational attainment, and healthcare access are crucial, quantifying the independent, net effect of each variable after accounting for their high levels of inter-correlation remains a complex statistical challenge.

The dataset utilized in this study captures cross-sectional data from 183 countries in 2015, providing a clear snapshot of vast global disparities in longevity. The broad range of Life Expectancy observed—from a minimum of 51 years to a maximum of 88 years (Table 3.1)—underscores the urgent need to identify the most potent determinants. This analysis is motivated by the necessity to move beyond simple correlation and provide robust, interpretable regression coefficients that policy-makers can use to prioritize resource allocation, particularly in developing nations where resources are scarce.

1.2 Problem Statement and Research Objective

The primary analytical problem addressed in this report is to identify and quantify the distinct contribution of several closely related socioeconomic and public health metrics to Life Expectancy, while rigorously controlling for the effects of multicollinearity that commonly plague such observational data.
The overarching objective of this research is twofold:

- To develop a parsimonious and statistically significant Multiple Linear Regression (MLR) model that accurately predicts Life Expectancy, using a minimal, yet highly explanatory, set of socioeconomic and health-related predictors.
- To interpret the resulting partial regression coefficients ($\hat{\beta}$) to determine the precise magnitude and direction of the independent impact of each predictor on Life Expectancy, thereby providing actionable, evidence-based insights for global health policy.

1.3 Report Structure

This report is structured into six comprehensive chapters designed to move systematically from data exploration to policy recommendations:

- **Chapter 1: Introduction** provides the background, motivation, and objectives of the study.
- **Chapter 2: Methodology** details the Multiple Linear Regression framework, including assumptions, coefficient interpretation (especially for log-transformed variables), and the criteria used for model selection, such as R^2_{adj} and Variance Inflation Factor (VIF) analysis.
- **Chapter 3: Data Description and Exploratory Analysis** presents the characteristics of the initial dataset, covering uni-variate distributions, necessary data transformations, and the initial bi-variate correlation structure.
- **Chapter 4: Statistical Model Development** outlines the steps for model fitting, including the strategy for resolving multicollinearity and the process of backward stepwise selection leading to the final model specification.
- **Chapter 5: Results** provides the empirical findings, including the overall model fit and a detailed interpretation of all seven significant predictor coefficients, quantifying their independent effects.
- **Chapter 6: Conclusion and Recommendations** synthesizes the statistical findings, offers evidence-based policy recommendations derived from the coefficients, and addresses the study's limitations and scope for future work.

Chapter 2

Methodology

2.1 Multiple Linear Regression Framework

Multiple Linear Regression (MLR) is the fundamental statistical technique employed in this study to model the linear relationship between a single continuous response variable (Life Expectancy) and multiple predictor variables. This framework is indispensable for quantifying the independent influence of each determinant while controlling for the effects of others. The general form of the MLR model is mathematically defined as:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + \epsilon_i$$

where Y_i represents the expected outcome for observation i, β_0 is the intercept, β_k are the partial regression coefficients, X_{ik} are the predictor variables, and ϵ_i is the random error term for observation i.

We expect an almost normal distribution of ϵ_i with mean zero and a constant variance, i.e,

$$\epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

2.1.1 Assumptions of Multiple Linear Regression

For the coefficient estimates to be unbiased, efficient, and for hypothesis tests to be valid, the MLR model relies on several core assumptions:

1. **Linearity:** The relationship between the predictors (X) and the response (Y) is correctly specified as linear.
2. **Normality:** The errors (ϵ_i) are normally distributed.
3. **Independence:** The errors(ϵ_i) are mutually independent to each others.
4. **Homoscedasticity:** The variance of the error terms (ϵ_i) is constant across all levels of the predictor variables (constant variance).

2.1.2 Coefficient Estimation and Interpretation

The regression coefficients (β_k) are estimated using the **Ordinary Least Squares (OLS)** method. OLS works by finding the line of best fit that minimizes the sum of squared vertical

distances between the observed data points(y_i) and the predicted regression line($\hat{y}_i$).
The MLR model be written in matrix form as:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

where

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \qquad \mathbf{X} = \begin{bmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1p} \\ 1 & X_{21} & X_{22} & \cdots & X_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{np} \end{bmatrix}, \qquad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}.$$

The vector of estimated coefficients, $\hat{\boldsymbol{\beta}}$, is obtained via the OLS closed-form solution:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}.$$

Each estimated coefficient $\hat{\beta}_k$ represents the expected change in the response variable (Y) associated with a one-unit increase in the predictor X_k, holding all other predictors constant.

When a predictor variable is logarithmically transformed (e.g., using $\log(X)$), the OLS coefficient β_k measures the sensitivity of the response variable Y to ***relative*** changes in the original variable X.

A relative change of $p\%$ in X is defined as:

$$p = \left(\frac{X_{\text{new}} - X_{\text{old}}}{X_{\text{old}}} \right) \times 100.$$

Such a change corresponds to the transformation:

$$\Delta \log(X) = \log(X_{\text{new}}) - \log(X_{\text{old}}) = \log\Big(1 + \frac{p}{100}\Big).$$

Since the regression model implies

$$\Delta Y \approx \beta_k \, \Delta \log(X),$$

the expected change in the response variable due to a $p\%$ relative increase in the original variable is:

$$\boxed{\Delta Y = \beta_k \, \log\Big(1 + \frac{p}{100}\Big)}$$

This formula is exact for any percentage change, not just small ones. For small p, we can use the approximation:

$$\log\Big(1 + \frac{p}{100}\Big) \approx \frac{p}{100},$$

which yields the commonly used rule of thumb:

$$\Delta Y \approx \frac{p}{100} \, \beta_k.$$

For a log-transformed predictor, the coefficient $\hat{\beta}_{\log(X)}$ approximately represents the change in Y associated with a 1% change in the original variable X.

2.1.3 Hypothesis Testing

Hypothesis testing is used to assess both the overall validity of the regression model and the contribution of each individual predictor. These evaluations are typically carried out through the F-test and the t-test, respectively.

Overall Model Significance

The overall explanatory power of the regression model is evaluated using the **F-test**. This test examines whether the set of predictors, taken together, provides a meaningful improvement in explaining the variation in the response variable.

The null and alternative hypotheses are:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0, \qquad H_1 : \text{At least one } \beta_k \neq 0.$$

Under the null hypothesis, the model does not outperform a model with only an intercept. A low p-value (commonly $p < 0.05$) indicates that the predictors collectively contribute significantly to explaining the variability in Y, and thus the model provides a statistically significant overall fit.

Significance of Individual Coefficients

The contribution of each predictor is assessed using the **t-test**. This test evaluates whether a specific regression coefficient differs significantly from zero after controlling for all other predictors in the model. For a given coefficient β_k, the hypotheses are:

$$H_0 : \beta_k = 0, \qquad H_1 : \beta_k \neq 0.$$

A low p-value associated with the t-statistic indicates that predictor X_k has a statistically significant independent effect on the response variable Y. In other words, there is evidence that changes in X_k are associated with changes in Y, even after accounting for the influence of other predictors in the model.

2.2 Model Selection Criterion

A parsimonious regression model explain a larger portions of the variations of outcome variable using a small subset of predictors. In this section, we discuss several important tools and techniques that we used to select an appropriate parsimonious regression model. These include stepwise regression, Adjusted R^2, and methods for diagnosing multicollinearity such as the Variance Inflation Factor (VIF).

2.2.1 Stepwise Regression

Stepwise regression is a variable–selection technique that iteratively adds or removes predictors based on predefined criteria such as the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), or significance levels(α). In this project, we primarily applied the **backward stepwise selection** method. **Backward Stepwise Regression:** Backward elimination begins with the *full model* that includes all candidate predictors. Then, at each

step, the least significant variable (often the one with the highest p-value) is removed. The reduced model is re-estimated, and the process continues until all remaining predictors satisfy the selection criterion. This approach is advantageous when the initial model contains many variables, as it systematically reduces complexity while preserving explanatory power.

2.2.2 Adjusted R^2

The Adjusted R^2 is a modified version of the ordinary coefficient of determination (R^2) that adjusts for the number of predictors in the model. Unlike R^2, which never decreases as more predictors are added, the Adjusted R^2 penalizes unnecessary variables and therefore provides a more reliable measure for model comparison.

The Adjusted R^2 for a linear regression model with n observations and k predictors is defined as:

$$R^2_{adj} = 1 - \frac{(1 - R^2)(n - 1)}{n - k - 1}$$

where SSE is the sum of squared errors and SST is the total sum of squares.

A larger Adjusted R^2 indicates a better-fitting model after accounting number of predictors added in the model. When comparing multiple models with different numbers of predictors, the model with the highest Adjusted R^2 is generally preferred.

2.2.3 Multicollinearity and VIF

Multicollinearity occurs when two or more predictor variables in a regression model are highly correlated. This can inflate standard errors, reduce the precision of coefficient estimates, and make it difficult to assess the individual effect of each predictor. One common diagnostic tool for multicollinearity is the Variance Inflation Factor (VIF).

For a predictor X_j, the Variance Inflation Factor is defined as:

$$\text{VIF}_j = \frac{1}{1 - R^2_j},$$

where R^2_j is the coefficient of determination from regressing X_j on all other predictors.

A VIF value of 1 indicates no multicollinearity, values between 5 and 10 indicate moderate multicollinearity, and values greater than 10 suggest severe multicollinearity that may require corrective action (such as removing or combining predictors).

Chapter 3

Data Description and Exploratory Analysis

3.1 Data Description

The dataset utilized in this analysis captures key public health, social, and economic indicators for 183 countries during the year 2015. The data were compiled from the Kaggle database, drawing primarily from established international sources such as the World Health Organization (WHO) and the United Nations (UN). The initial dataset is comprehensive, comprising 15 variables in total: the response variable, the average life expectancy, and 13 potential predictor variables, and an identifier variable. For the purpose of structured analysis, these predictors have been broadly categorized into two major thematic groups: health-related variables and socioeconomic and demographic variables.

3.1.1 Outcome Variable

- **Life Expectancy (`le`)**: As the designated response variable, Life Expectancy is defined as the average number of years a person is expected to live, assuming present mortality patterns continue. It is the core variable being modeled and represents the overarching measure of a country's success in providing health and socioeconomic stability.

3.1.2 Socioeconomic and Demographic Variables

This group consists of one identifier variable and five indicators that capture the fundamental economic and social structure of the countries, influencing overall health capacity and resource distribution:

- **Country (`Country`)**: A categorical identifier for the observation unit, representing the country name.
- **Country Status (`status`)**: A nominal categorical variable indicating whether the country is officially classified as Developed or Developing. In the regression, this is converted to a binary indicator (`statusDeveloping`).
- **Income Composition Index (`inComposition`)**: An index between 0 to 100 reflecting the distribution and equality of income within the country. A higher score represents

better income equality within the country. It's a key socioeconomic determinant associated with access of healthcare facility. Unequal income composition is often correlated with unequal access to healthcare.

- **Schooling** (`schooling`): The average number of years of schooling completed by adult populations who are 25 years old or higher. Education is a key socioeconomic determinant strongly correlated with health literacy, access to employment, and quality of life.
- **GDP per Capita** (`gdp`): Gross Domestic Product per capita, measured in US Dollars. It serves as the primary measure of a country's economic production and wealth, directly influencing national resource availability for public services.
- **Population** (`population`): The total population size of the country in the observed year. This variable accounts for the scale of the population when assessing health or economic indicators.

3.1.3 Health-Related Variables

This group consists of eight indicators designed to capture the overall health status of a country's population, as detailed below:

- **Adult Mortality Rates** (`adMortality`): Measures the percentage of adults who die before reaching the age of 60. As a fundamental measure of health stability and infrastructure strength, a consistently higher rate is generally an indicator of a weaker public health system and higher disease burden.
- **Early Childhood Thinness** (`thin9`): Refers to the percentage of children aged 5 to 9 years old classified as underweight. Underweight is technically defined as having a weight-for-age that is less than two standard deviations below the median weight of the international reference population for this age group. This variable focuses on children entering school age. A high rate indicates poor early-life nutrition that is likely to impact cognitive development, school performance, and disease tolerance.
- **Adolescent Thinness** (`thin19`): Refers to the percentage of children aged 10 to 19 years old classified as underweight. This variable covers the broad span from infancy through late adolescence, providing an indicator of chronic deficits that affect growth and development during key years of physical maturity.
- **HIV Death Rates** (`HIV`): Indicates the percentage of children who die before their fifth birthday specifically due to AIDS-related illness. This variable is essentially the Child Mortality Rate component that is attributable to the HIV epidemic. It serves as a strong indicator of how well a country is implementing its programs to prevent the mother-to-child transmission of HIV.
- **Hepatitis B Immunization Deficit** (`hepB`): Represents the percentage of 1-year-olds who have been **not** immunized against Hepatitis B. A higher rate indicates poor vaccination coverage.

- **Polio Immunization Deficit** (`polio`): Represents the percentage of 1-year-olds who have been **not** immunized against Polio. A higher rate indicates poor vaccination coverage.
- **Diphtheria Immunization Deficit** (`diphtheria`): Represents the percentage of 1-year-olds who have been **not** immunized against Diphtheria. A higher rate indicates poor vaccination coverage.
- **Measles Cases** (`measles`): Refers to reported cases of measles per 100 population.

3.2 Exploratory Data Analysis

3.2.1 Uni-variate Analysis

Table 3.1 presents the central tendency, spread, and range for all 16 variables utilized in the initial exploratory analysis, including the raw form of the response variable, Life Expectancy. Analyzing the minimum and maximum values confirms the necessity of the log-transformations applied to highly skewed economic variables like GDP and high-disparity public health metrics such as HIV Death Rate and Immunization Deficits. The table highlights the substantial variation in wealth and health outcomes across the 183 countries in the 2015 dataset.

The distribution of the response variable, Life Expectancy (`le`), provides a clear overview of the

Table 3.1: Descriptive Statistics for all variables

Variable	Min	Q_1	Median	Mean	Q_3	Max
`le` (Life Expectancy)	51.00	65.75	73.90	71.62	76.95	88.00
`inComposition`	0.35	0.58	0.77	0.69	0.79	0.95
`schooling`	4.90	11.10	13.10	12.94	14.85	20.40
`gdp`	13.68	1152.31	2916.23	6508.81	5881.99	66346.52
`adMortality`	1.00	7.44	13.8	15.29	21.3	48.4
`HIV`	0.01	0.01	0.01	0.66	0.40	0.93
`thin9`	0.10	1.50	3.40	4.56	6.40	27.30
`thin19`	0.10	1.50	3.50	4.52	6.50	26.70
`hepB`	1.00	3.00	17.05	7.00	19.00	94.00
`polio`	1.00	3.00	16.79	7.00	17.00	95.00
`diphtheria`	1.00	3.00	16.37	7.00	16.50	94.00
`measles`	0.00	0.00	0.02	1.50	0.20	90.3
`population`	2966	604820	2076086	9076238	6475798	258162113

range and central tendency of longevity across the 183 countries in the 2015.Numerically, the average life expectancy is 71.62 years, with the middle fifty percent of countries falling between 65.75 years and 76.95 years. The overall range is substantial, spanning from a minimum of 51 years to a maximum of 88 years, reflecting the vast global disparities in health outcomes and development. Graphically, the histogram in Figure 3.1 suggests a distribution that is slightly left-skewed, meaning the tail extends towards the lower life expectancies. However, the distribution is generally close to normal around the central mean. The box plot in Figure 3.1 confirms this, showing a relatively compact inter-quartile range. This near-normal distribution

is beneficial for linear regression modeling, as the normality assumption primarily applies to the residuals, but a normally distributed response variable often aids in achieving this.

The categorical variable Country Status (`status`) partitions the 183 countries into two

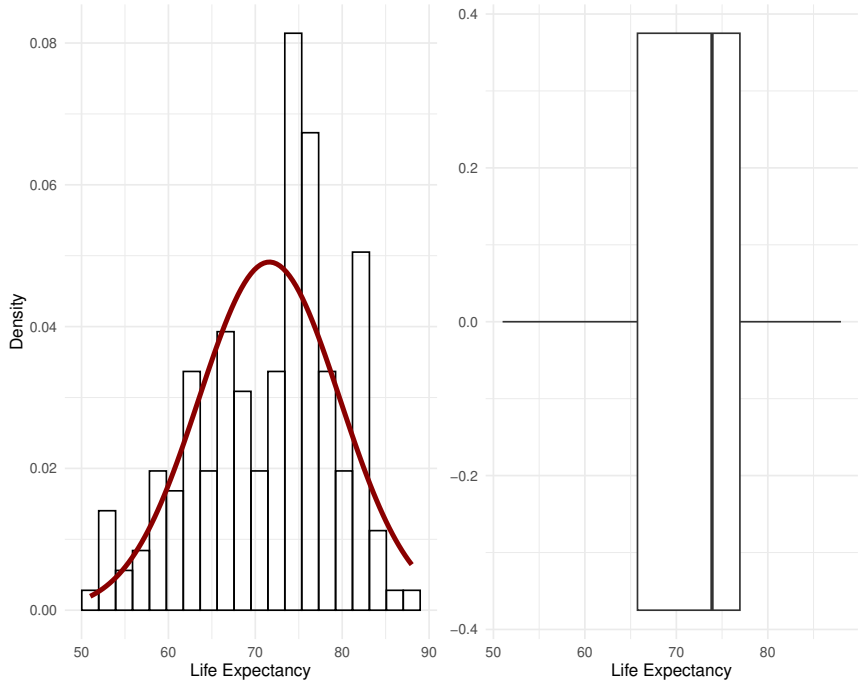


Figure 3.1: Distribution of Life Expectancy

categories: Developed and Developing. The majority of observations fall under the Developing category, totaling 151 countries or 82.5% of the dataset. Conversely, the Developed status accounts for only 32 countries, representing 17.5% of the total sample.(Figure 3.2)

The Income Composition Index (`inComposition`), an index ranging from 0 to 1, reflects the distribution and equality of income within a country, with higher scores indicating better income equality. The variable is broadly distributed around a mean of 0.6934 and a median of 0.7230. The inter-quartile range is relatively wide, spanning from 0.5755 to 0.7945, demonstrating substantial global variation in income equality. This range is anchored by a minimum value of 0.3470 (exemplified by the Central African Republic) and a near-perfect maximum of 0.9480 (represented by Norway). Unlike several other socioeconomic variables, the raw Income Composition Index exhibits a stable, relatively symmetric distribution(Figure 3.3) and, crucially, demonstrates a significant and strong positive linear association with Life Expectancy in exploratory analysis. Due to this favorable linearity and distribution, the raw variable was directly employed in the multiple regression model without any transformation.

The average number of years of schooling completed by the adult population (`schooling`) demonstrates a distribution that is relatively centralized and symmetric, which is ideal for regression modeling. The mean years of schooling stands at 12.94 years, closely aligned with the median of 13.10 years, suggesting a minimal skew. The majority of countries have adult

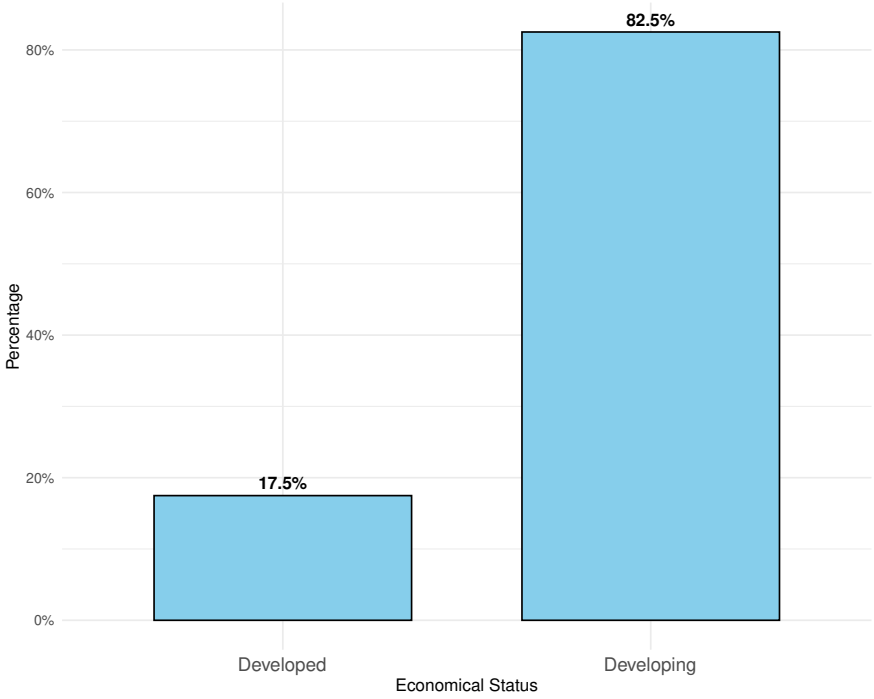


Figure 3.2: Distribution of Economic Status

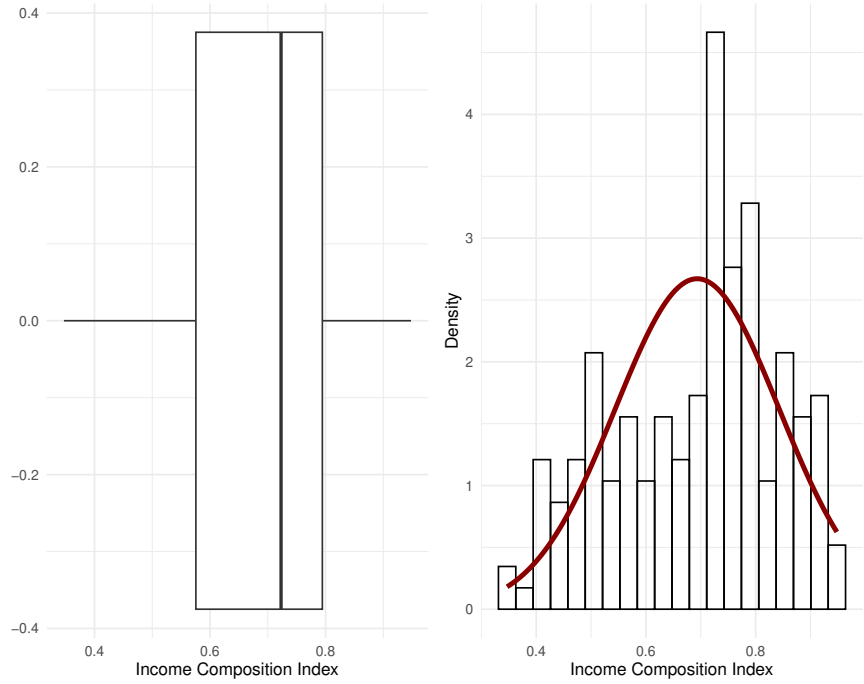


Figure 3.3: Distribution of Income Composition Index

populations completing between 11.10 years and 14.85 years of education. The overall range is broad, extending from a minimum of 4.90 years to a maximum of 20.40 years, reflecting significant global variation in educational attainment. Crucially, as confirmed by Figure 3.4, the variable exhibits no discernible outliers or extreme values, which increases confidence in its stability within the model. The centralized and well-behaved nature of the `schooling` variable makes it an excellent candidate for direct inclusion in the multiple linear regression model.

GDP per Capita (`gdp`) is the primary economic indicator in the model and exhibits an

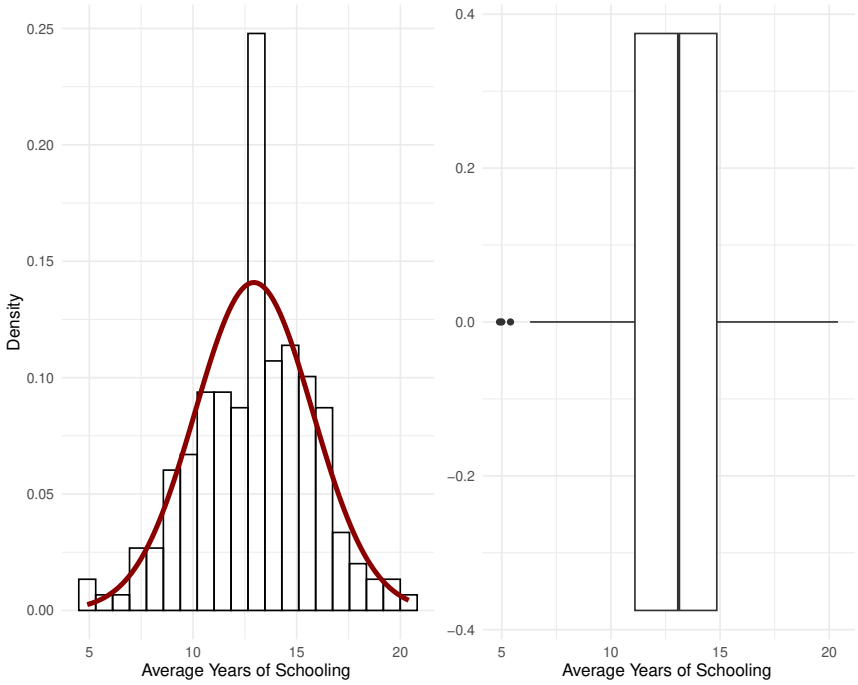


Figure 3.4: Distribution of Years of Schooling

extremely skewed distribution, reflecting the massive disparity in global wealth. The distribution is strongly right-skewed, evidenced by the large difference between the median GDP of $2,916.23 and the significantly higher mean of $6,508.81. The wealth range is immense, spanning from a minimum of $33.68 to a maximum of $66,346.52. As shown in Figure 3.5 (upper panel), the raw `gdp` variable displays this heavy positive skew, which, when coupled with its non-linear relationship with Life Expectancy in exploratory bi-variate analysis, would violate key regression assumptions. To rectify this, a logarithmic transformation was applied. The resulting `log(gdp)` variable successfully achieved a much cleaner, more symmetric distribution (Figure 3.5, lower panel) and, critically, established the necessary moderate positive linear relationship with Life Expectancy, justifying its use in the final multiple regression model. Therefore, all subsequent model results involving this predictor refer to the log-transformed GDP per capita.

The Adult Mortality Rate (`adMortality`) is a critical health determinant in this analysis. The distribution exhibits a clear right-skew, with the mean rate of 15.29% being slightly pulled upward from the median of 13.80%. The majority of countries fall within a range defined

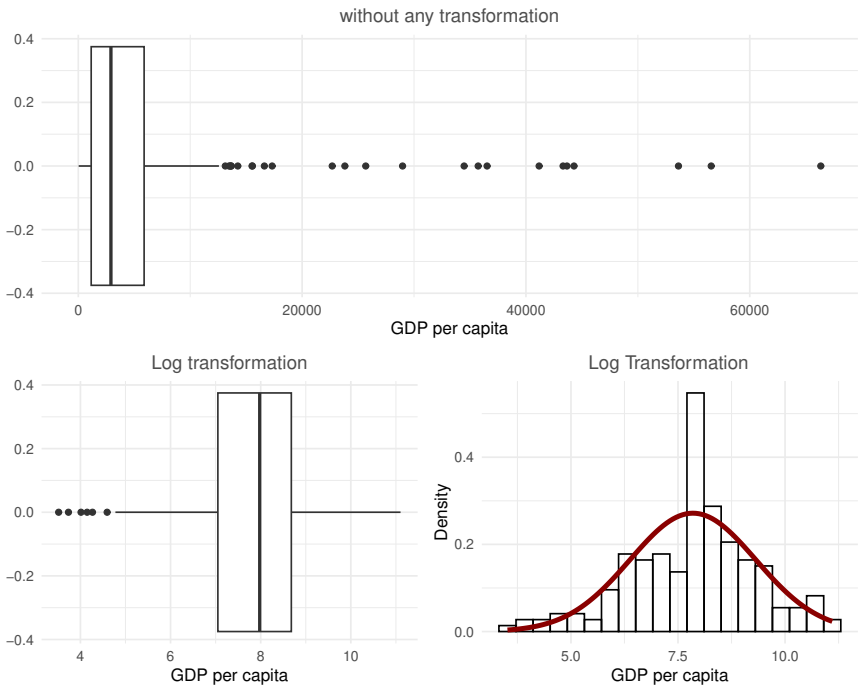


Figure 3.5: Distribution of GDP Per Capita

by the quartiles of 7.40% and 21.30%, but the overall spread is significant, extending up to a maximum of 48.40%. This maximum value, belonging to the country of Lesotho, stands out as a pronounced outlier, with the next highest rate being substantially lower at 41.30%. As depicted in Figure 3.6, while the uni-variate distribution is skewed, exploratory bi-variate analysis (to be shown later) demonstrates a strong, clear inverse linear relationship between the raw Adult Mortality Rate and Life Expectancy. Therefore, despite the skewness, the untransformed variable is retained for the initial multiple linear regression model, and the potential influence of the Lesotho outlier will be formally assessed in the diagnostics phase of the modeling process.

The HIV Death Rate variable (`HIV`) is characterized by an extremely non-symmetric distribution. The vast majority of countries report minimal rates, resulting in a high concentration of data points at the floor, which is reflected by the first quartile and median both equaling 0.01%. However, the mean of 0.066% is strongly pulled by the maximum rate of 0.93%, indicating a severe positive skew. As seen in Figure 3.7 (left panel), this highly skewed raw distribution that lacked the linearity with outcome variable required for the multiple regression model. To correct this non-linearity, two transformations were tested: the logarithmic and the log-of-log transformation. Although the log-of-log transformation achieved a slightly more normal-looking distribution, the logarithmic transformation (right panel of Figure 3.7) was ultimately selected as it resulted the strongest and most suitable inverse linear association with the Life Expectancy response variable. This choice sacrifices perfect uni-variate normality for the sake of better model fit and interoperability of the linearity assumption. Therefore, the log-transformed HIV Death Rate is the form utilized in the final model.

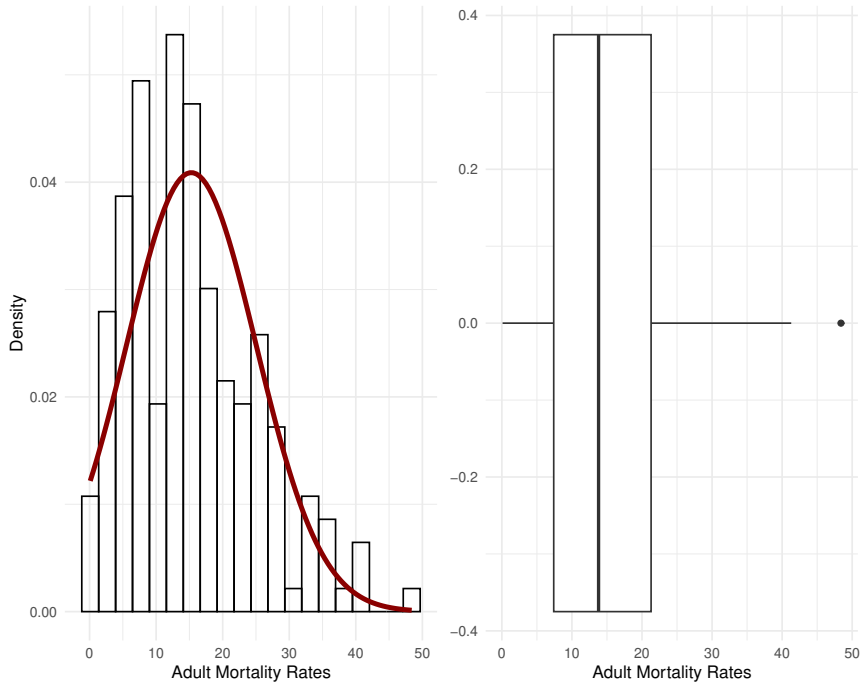


Figure 3.6: Distribution of Adult Mortality Rates

The distribution of Early Childhood Thinness indicator(`thin9`) measures the percentage of children aged 5 to 9 classified as underweight, and its distribution is significantly impacted by extreme values. The data is heavily concentrated at lower thinness rates, with the median at 3.40% and the mean pulled higher to 4.56% due to a severe positive skew. The range spans from a minimum of 0.10% to a maximum of 27.30%.A crucial feature of this distribution, evident in Figure 3.8 (upper panel), is the presence of a distinct cluster of high-leverage outliers, notably India, Pakistan, Bangladesh, and Afghanistan (a region heavily impacted by malnutrition). These high values disrupt the linear relationship with Life Expectancy.Consequently, a logarithmic transformation was applied to `thin9`. This transformation proved effective in stabilizing the variance and establishing the necessary linear association required for the regression model. As confirmed by Figure 3.8 (lower panel), the use of the log-transformed Early Childhood Thinness variable addresses the issues of non-linearity and skewness, thus allowing the final model to reliably estimate the inverse effect of early childhood malnutrition on Life Expectancy.

The Adolescent Thinness variable (`thin19`) shows a statistical pattern nearly identical to its counterpart, Early Childhood Thinness (`thin9`). The distribution is heavily right-skewed, with the mean (4.52%) being pulled higher than the median (3.50%). The spread ranges from a minimum of 0.10% to a maximum of 26.70%. Given this strong positive skew, and to establish linearity with Life Expectancy, a necessary logarithmic transformation was applied. As illustrated in Figure 3.9, the resulting `log(thin19)` variable possesses the necessary symmetrical distribution for modeling purposes. Crucially, the extreme similarity in both distribution and overall values leads to an observed bi-variate correlation of $\mathbf{R = 0.97}$ between `thin19` and `thin9`. This extremely high correlation indicates significant multicollinearity,

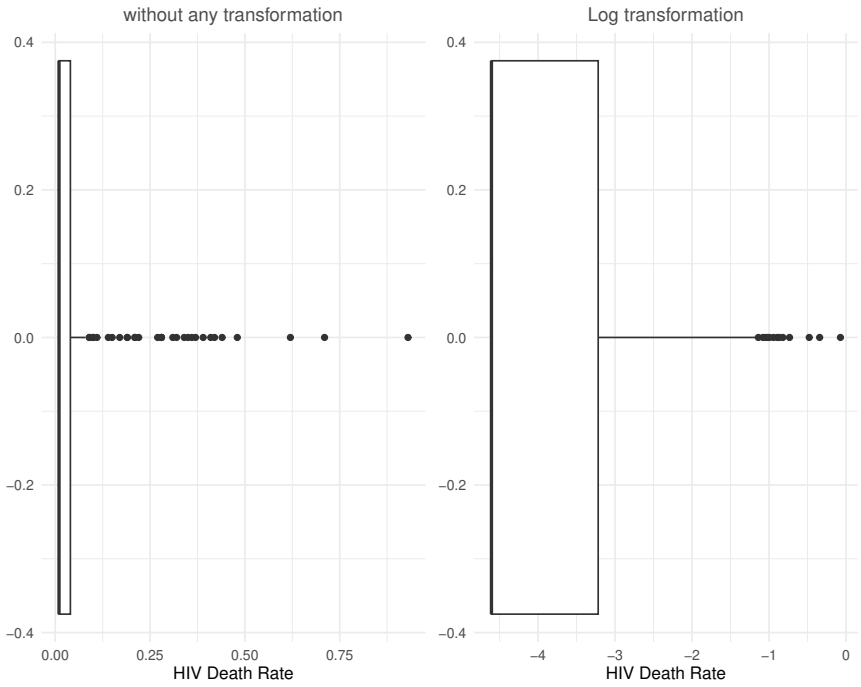


Figure 3.7: Distribution of Adult Mortality Rates

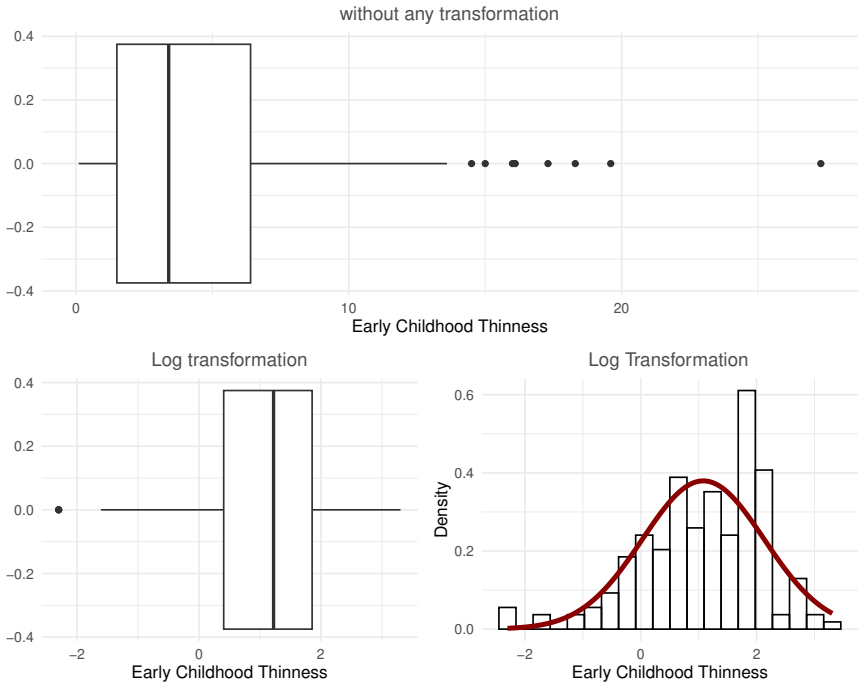


Figure 3.8: Distribution of Early Childhood Thinness

suggesting that only one of these two variables should be retained in the final multiple regression model to ensure coefficient stability (a decision that will be confirmed using Variance Inflation Factors).

The Hepatitis B Immunization Deficit variable (`hepB`), defined as the percentage of

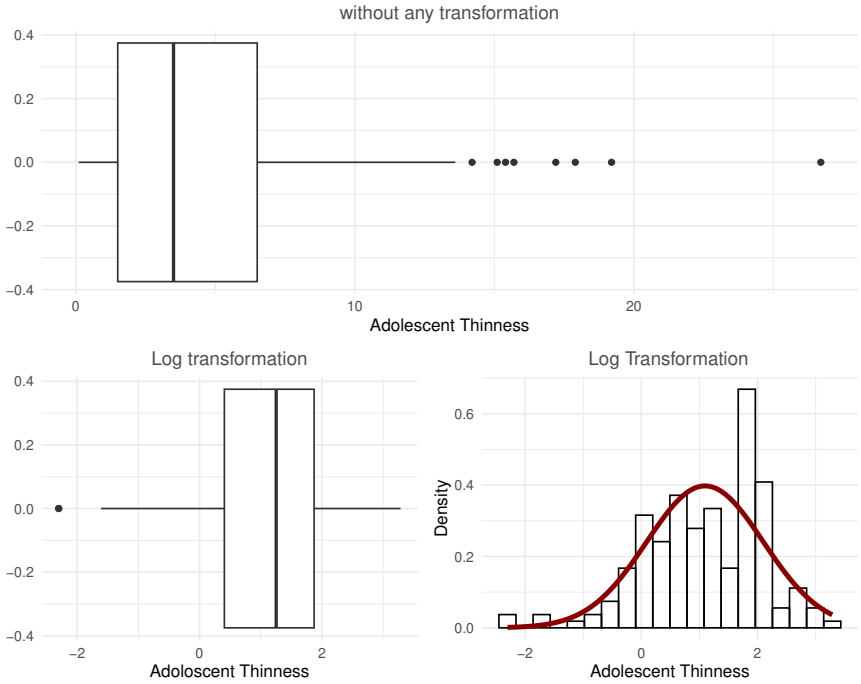


Figure 3.9: Distribution of Adolescent Thinness

one-year-olds who have **not** received the Hepatitis B vaccine, reinforces the pattern observed in other immunization indicators. The distribution is highly right-skewed, with the median deficit at 7.00% and the mean pushed to 17.05%. The data ranges from a minimum deficit of 1.00% to a maximum of 94.00%. This wide spread is driven by nine high-deficit outliers (e.g., Congo and Haiti), where deficit rates surpass **90**%. As shown in Figure 3.10, the raw data's skewness necessitated a logarithmic transformation to achieve linearity with Life Expectancy. However, due to the extremely high correlation observed between the Hepatitis B Deficit and both the Diphtheria and Polio Deficits, this variable was ultimately excluded from the final regression model during the variable selection process. Its strong association with the response variable is effectively captured by the retained, highly correlated Polio Immunization Deficit, allowing us to mitigate serious multicollinearity issues.

The variable Polio Immunization Deficit (`polio`) represents the percentage of one-year-olds who have **not** been immunized against Polio. This metric is strongly right-skewed, indicating that while most countries have very low deficits, the overall average of 16.79% is notably high due to a significant tail of high values. The distribution is highly concentrated, with the median deficit at 7.00% and the 1st quartile at just 3.00%. This disparity is driven by 11 pronounced outliers, where the deficit rates exceed 90% (e.g., Kiribati and Tonga). These observations, representing small and often poor countries struggling with vaccine delivery, create a severe deviation from the main distribution. As illustrated in Figure 3.11 (upper panel), the raw

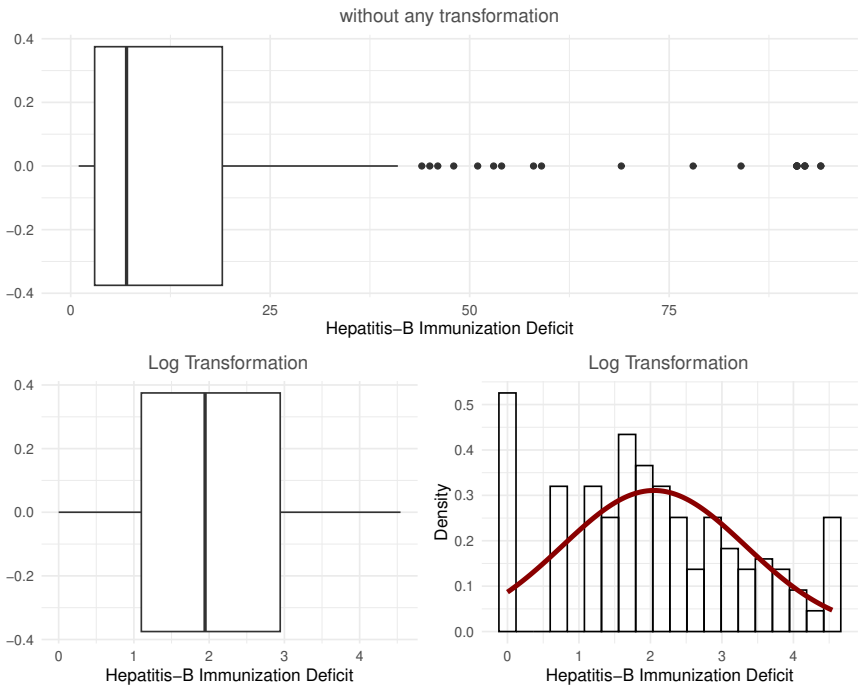


Figure 3.10: Distribution of Hepatitis-B Immunization Deficit

variable's skewness and outlier presence led to a non-linear relationship with Life Expectancy. Consequently, a logarithmic transformation was applied. The resulting log-transformed Polio Deficit (right panel) provided the necessary variance stabilization and, critically, established a moderate inverse linear relationship with Life Expectancy. The transformed variable is therefore included in the final regression model.

The Diphtheria Immunization Deficit variable (`diphtheria`), representing the percentage of one-year-old not immunized against Diphtheria-Pertussis-Tetanus (DPT), exhibits a distribution pattern that is strikingly similar to the Polio Deficit. The variable is characterized by a strong right-skew, with the median deficit at 7.00% and the average deficit rate at 15.37%. The minimum deficit is 1.00%, while the maximum reaches 94.00%. This distribution features a distinct group of nine outliers—small and developing countries like Congo and Mozambique—where the deficit exceeds 83%, highlighting severe gaps in routine immunization coverage. Given its pronounced skew and the strong, expected relationship with Life Expectancy, a logarithmic transformation was applied (the log-transformed plot is shown in Figure 3.12 alongside the raw data). However, due to its extremely high correlation with the already selected Polio Immunization Deficit and other health indicators, the Diphtheria Deficit variable was later excluded during the variable selection phase to mitigate issues of multicollinearity in the final multiple regression model.

The Measles Cases variable (`measles`), which records reported cases per 100 population, required initial data cleaning where two typographical errors were corrected by imputation using the median value (**0.0170**). The variable exhibits an extremely positive-skewed distribution, characterized by an abundance of zero values. Numerically, the median number of cases is nearly zero (**0.0170**), while the mean is substantially higher at **0.7777**, driven by

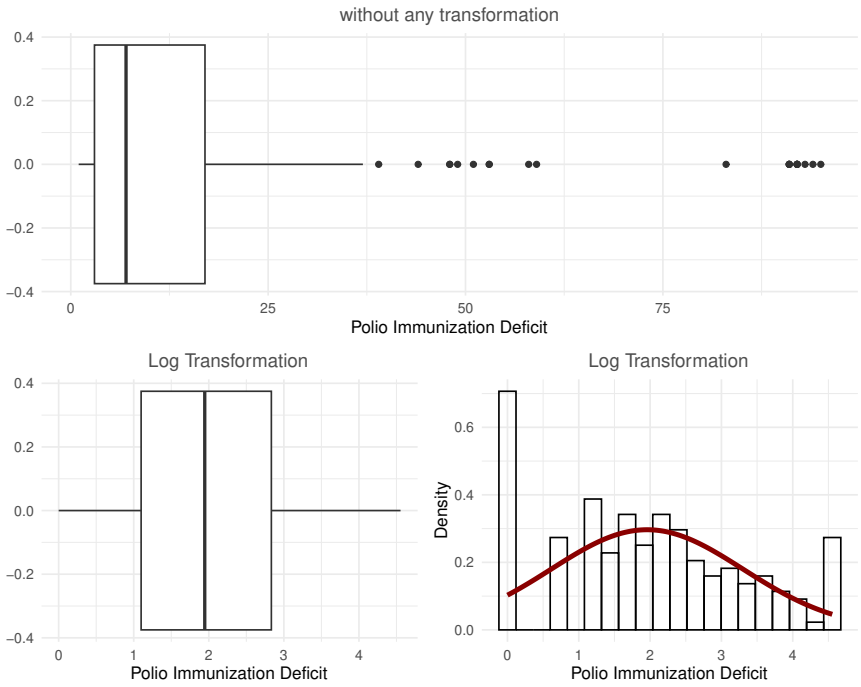


Figure 3.11: Distribution of Polio Immunization Deficit

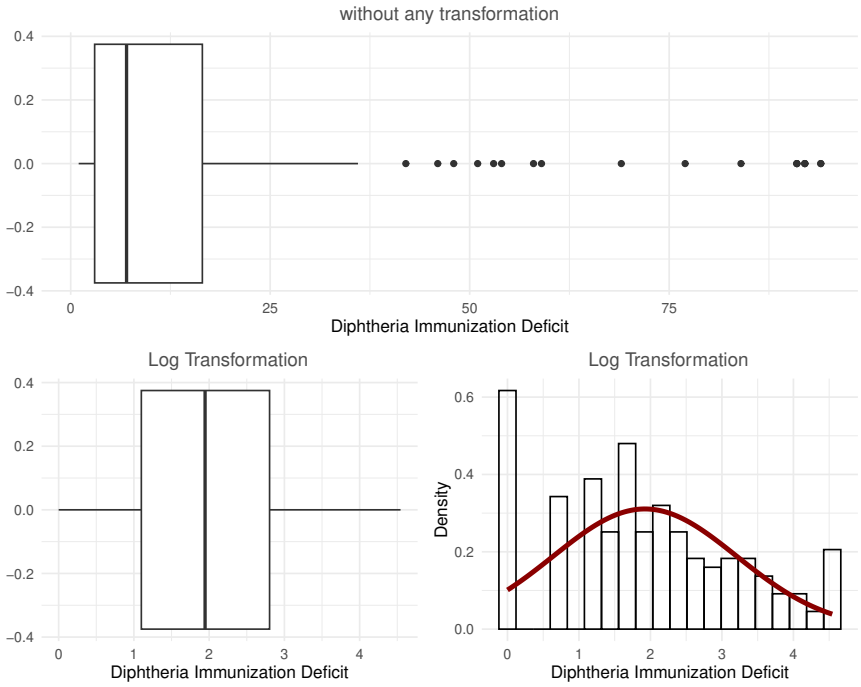


Figure 3.12: Distribution of Diphtheria Immunization Deficit

a long tail extending to a maximum of **20.3590**. Due to the zero-inflated nature of the data, a logarithmic transformation utilizing an offset, `log(measles` $+ \mathbf{0.5}$), was applied to stabilize variance and allow for regression analysis. As depicted in Figure 3.13 (right panel), the transformation successfully normalized the distribution. However, exploratory analysis showed only a weak negative linear association with Life Expectancy, indicating that while outbreaks are detrimental, this variable was statistically less relevant than the stronger indicators (e.g., `adMortality` or `HIV`) and was therefore excluded during the final variable selection to maintain a parsimonious model.

The Population variable (`population`), reflecting the total number of individuals in a country,

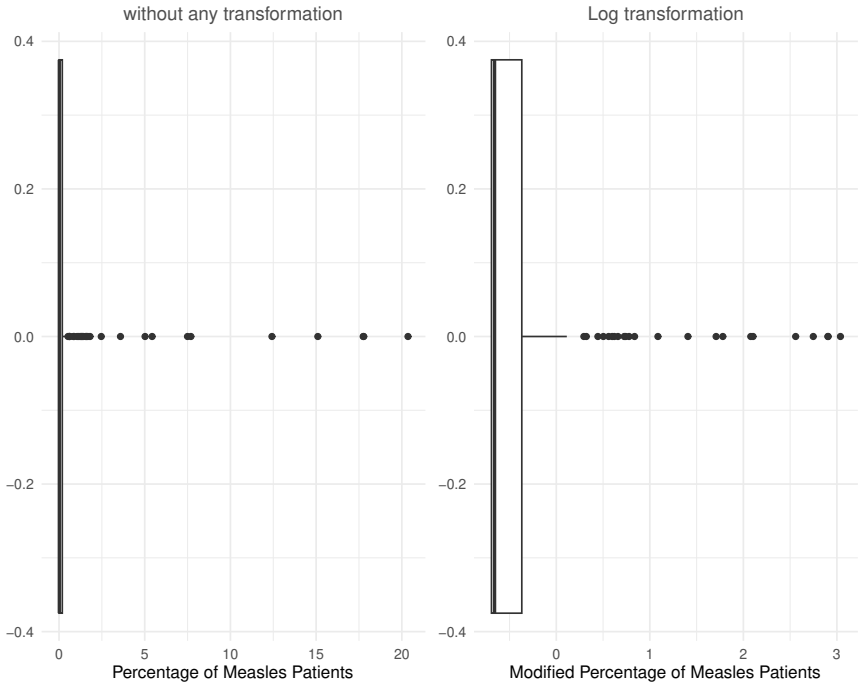


Figure 3.13: Distribution of Percentages of Measles Cases

exhibits extreme positive skewness, spanning from a minimum of 2,966 to a maximum of over 258 million. This massive scale difference, with the mean (9.7 million) far exceeding the median (20.7 million), necessitates treatment prior to modeling. A logarithmic transformation was successfully applied to normalize this highly skewed distribution, as illustrated in Figure 3.14 (right panel). However, exploratory analysis showed that neither the raw nor the transformed population variable held any significant linear association with Life Expectancy, demonstrating a correlation of approximately $\mathbf{R = -0.027}$. Therefore, Population was excluded from the final multiple regression model, confirming its intuitive status as a scale variable with negligible direct predictive power regarding public health outcomes.

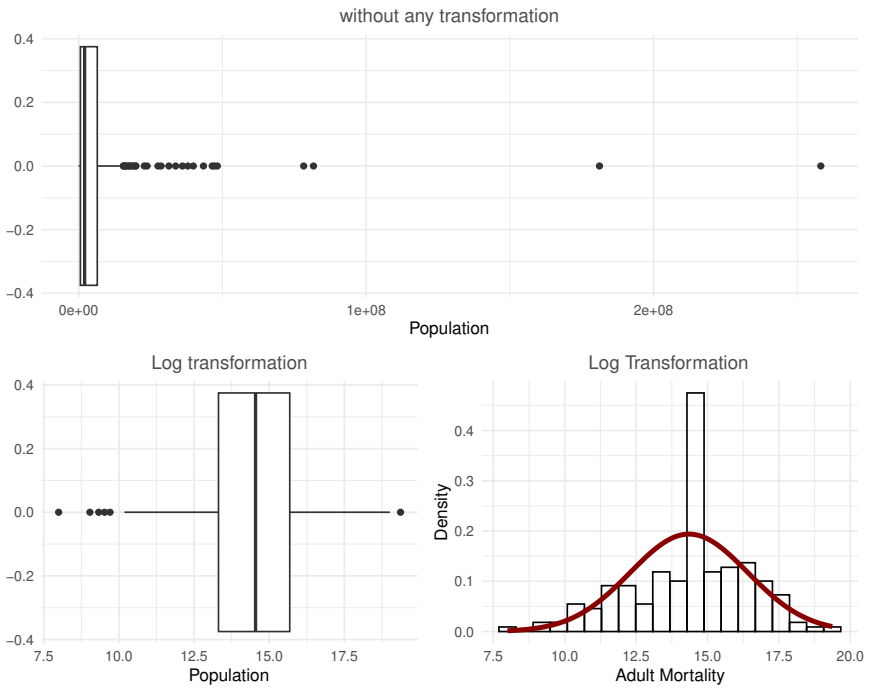


Figure 3.14: Distribution of Population

3.2.2 Bi-variate Analysis

Relationship between Economic Status and all variables

The first stage of the bi-variate analysis focused on exploring the relationship between the categorical variable, Country Status (`status`), and the mean distribution of the numerical health and socioeconomic indicators, as detailed in Table 3.2. A significant gap is evident across nearly all metrics based on a country's economic classification. Developed countries consistently exhibit more favorable outcomes, characterized by a higher mean Life Expectancy, greater average years of Schooling, higher GDP per capital, and a better income-inequality. Conversely, developing countries generally struggle with higher Adult Mortality Rates, elevated HIV Death Rates, a greater prevalence of malnutrition, and significantly larger immunization deficits. This pronounced difference in mean values confirms the strong predictive capacity of the Country Status variable, justifying its integral role as a primary predictor in the initial model specification.

Relationship between Life Expectancy and Socioeconomic Variables

The assessment of bi-variate relationships between Life Expectancy and the socioeconomic predictors confirms the expected positive correlations. The correlation matrix, displayed in Figure 3.15, reveals that all three primary socioeconomic indicators exhibit a strong, positive linear relationship with Life Expectancy. The Income Composition Index ($R = 0.86$) shows the highest individual correlation with Life Expectancy, closely followed by Years of Schooling ($R = 0.78$). Both demonstrate highly significant positive associations, meaning higher educational attainment and better income equality strongly predict longer life expectancy.

Table 3.2: Average of Life Expectancy and Predictor variables based on Economic Status

Variable	Developed Country	Developing Country
`le` (Life Expectancy)	80.70	69.69
`inComposition`	0.87	0.66
`schooling`	16.22	12.24
`gdp`	14745.12	4763.36
`adMortality`	7.48	16.93
`HIV`	0.01	0.07
`thin9`	1.20	5.27
`thin19`	1.26	5.22
`hepB`	9.44	18.67
`polio`	4.94	17.62
`diphtheria`	4.75	17.62
`measles`	0.12	1.80
`population`	9376299	9012648

The log-transformed GDP per Capita (`log(GDP)`) also maintains a strong positive relationship ($R = 0.49$), and the scatter plot confirms that the logarithmic transformation successfully linearized the relationship between wealth and longevity.

Relationship between Life Expectancy and Health Related Predictors

The analysis of the health-related predictors reinforces the strong influence of public health burdens on Life Expectancy. The key findings are highlighted in Figure 3.16. As expected, all health burden variables display a strong inverse linear relationship with Life Expectancy. The strongest predictor among this group is the raw Adult Mortality Rate (`adMortality`), which exhibits a profound negative correlation ($R \approx -0.80$). The log-transformed HIV Death Rate (`log(HIV)`) and the log-transformed Early Childhood Thinness (`log(thin9)`) also show robust negative associations ($R \approx -0.65$ and $R \approx -0.60$, respectively). This means that lower life expectancy is highly predictable by higher adult death rates and a greater prevalence of chronic malnutrition or AIDS-related mortality.

Possible Multicolinearity

The bi-variate analysis conclusively demonstrated significant linear dependence among several predictor variables, necessitating a rigorous variable selection procedure in Chapter 4 to ensure model reliability. The primary concerns identified relate to high redundancy across thematic clusters. Specifically, the socioeconomic redundancy between `inComposition` index and the `schooling`, evidenced by an extremely high correlation ($\mathbf{R = 0.92}$), suggests that these two variables should not be simultaneously included in the regression model. Furthermore, immunization redundancy is present due to the near-perfect correlations ($R > 0.90$) observed among the immunization deficit variables (`polio`, `diphtheria`, and `hepB`), where retaining more than one representative variable would introduce severe instability. Finally, malnutrition redundancy, highlighted by the high correlation ($\mathbf{R = 0.97}$) between `thin9` and `thin19`, likewise dictates the retention of only one measure of thinness prevalence. These critical findings transition the focus of the analysis from data exploration to formal model building.

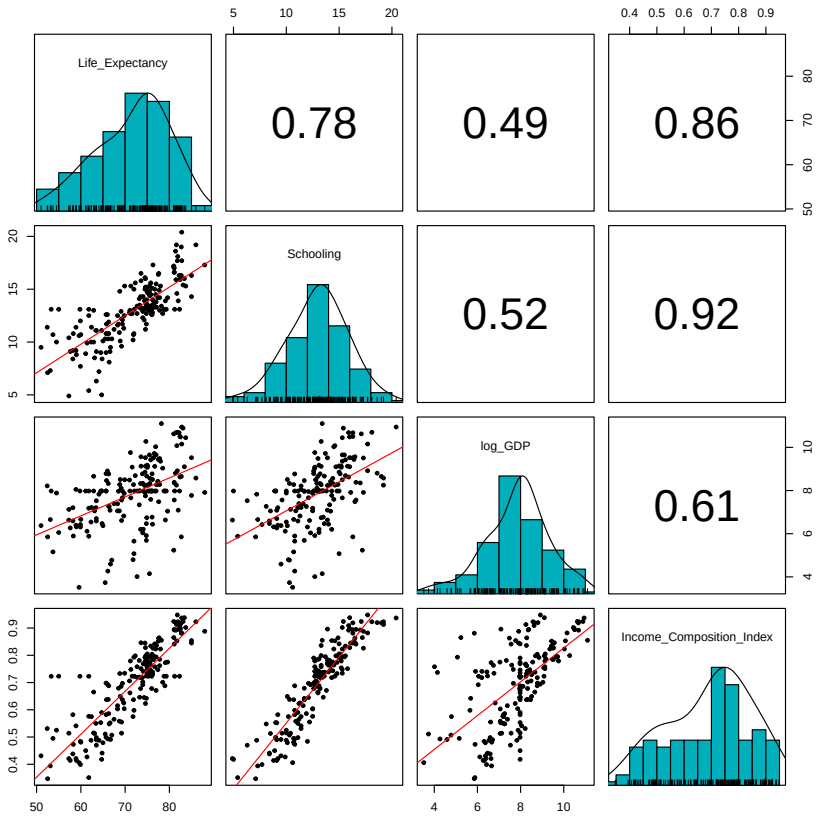


Figure 3.15: Correlation Matrix between Life Expectancy and Socioeconomic Predictors

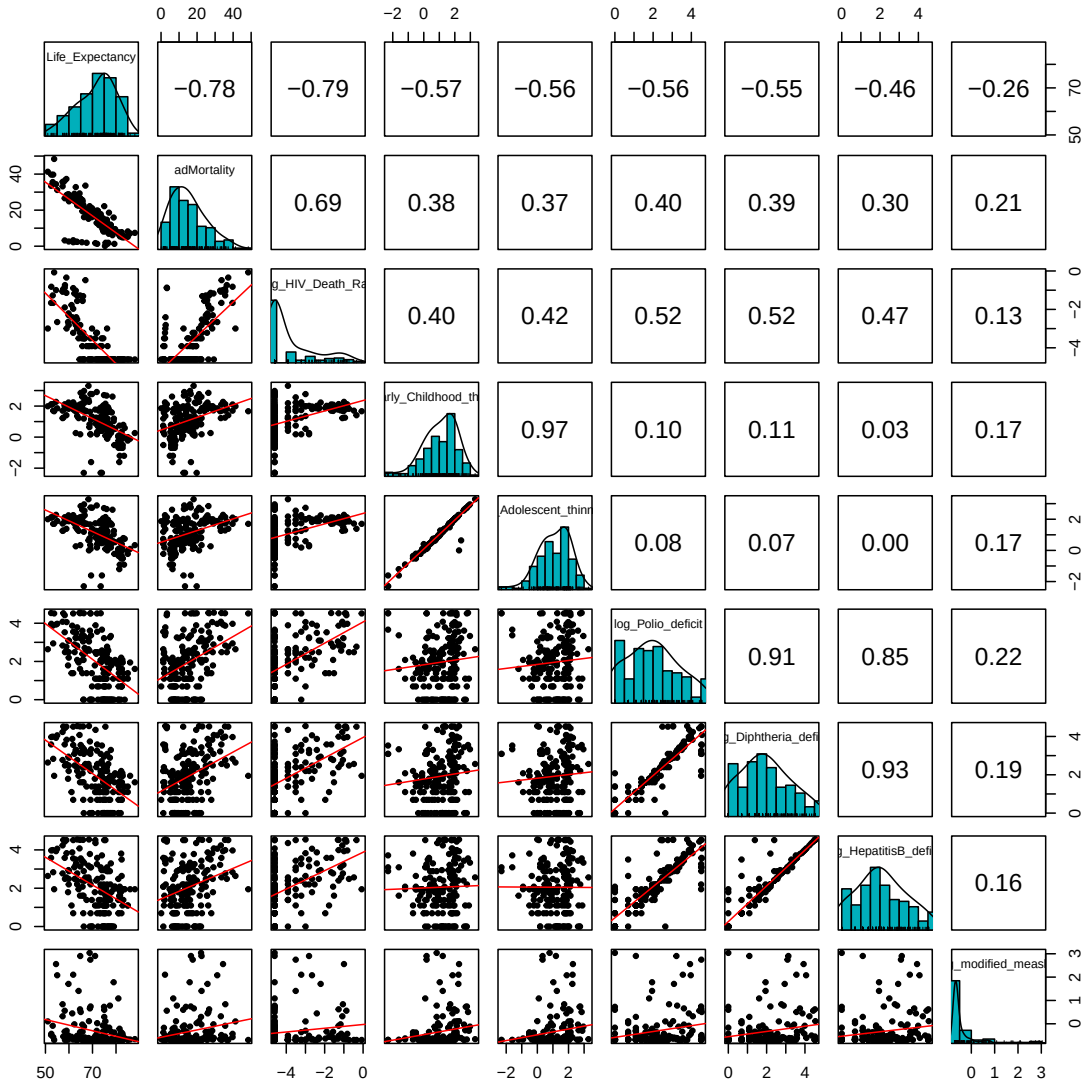


Figure 3.16: Correlation Matrix between Life Expectancy and Health related Predictors

In the next chapter, we will use Variance Inflation Factors (VIFs) to formally quantify the severity of these dependencies and systematically eliminate redundant variables, selecting the most parsimonious set of predictors that maintain high predictive power while ensuring coefficient stability.

Chapter 4

Statistical Model Development

4.1 Final Model Development

In this chapter, we describe the process used to develop a statistically valid and interpretable multiple regression model. The procedure consisted of

1. constructing an initial full model followed by
2. addressing multicollinearity and then
3. applying backward stepwise regression to remove insignificant variables.

Diagnostic plots were then used to assess the validity of the final fitted model.

4.1.1 Initial Full Model

We began with a full regression model containing all available transformed predictors from the dataset(Equation 4.1). The aim was to identify a parsimonious and statistically reliable model while maintaining higher interoperability and satisfying regression assumptions.

$$\begin{aligned}\hat{le} = \hat{\beta}_0 &+ \hat{\beta}_1\,\texttt{inComposition} + \hat{\beta}_2\,\texttt{schooling} + \hat{\beta}_3\,\log(\texttt{gdp}) + \hat{\beta}_4\,\log(\texttt{population}) \\ &+ \hat{\beta}_5\,\texttt{adMortality} + \hat{\beta}_6\,\log(\texttt{HIV}) + \hat{\beta}_7\,\log(\texttt{thin9}) + \hat{\beta}_8\,\log(\texttt{thin19}) + \hat{\beta}_9\,\texttt{polio} \\ &+ \hat{\beta}_{10}\,\log(\texttt{diphtheria}) + \hat{\beta}_{11}\,\log(\texttt{hepB}) + \hat{\beta}_{12}\,\log(\texttt{measles} + 0.5) + \hat{\beta}_{13}\,\texttt{status}.\end{aligned} \tag{4.1}$$

where,

$$\texttt{status} = \begin{cases} 0 & ;\ \text{Developed Country,} \\ 1 & ;\ \text{Developing Country.} \end{cases}$$

4.1.2 Assessment of Multicollinearity

Before performing variable elimination based on statistical significance, we examined multicollinearity using the Variance Inflation Factor (VIF). A VIF greater than **5** was treated as a threshold indicating potentially harmful collinearity.

At each step, the predictor with the highest VIF was removed, and the model was refitted so that all remaining VIF values fell below the acceptable limit. This iterative process was repeated three times:

- **Run 1:** Removed `thin19`
- **Run 2:** Removed `diphtheria`
- **Run 3:** Removed `inComposition`

After these removals, all VIF values were below 5(Table 4.1). At this point, the model no longer exhibited critical levels of multicollinearity, so we proceeded to the next stage.

Table 4.1: VIF table

`schooling`	2.54
log(`gdp`)	1.40
log(`population`)	1.09
`adMortality`	2.11
log(`HIV`)	2.56
log(`thin9`)	1.69
log(`polio`)	4.27
log(`hepB`)	3.90
log(`measles` + 0.5)	1.16
`status`	1.59

4.1.3 Backward Stepwise Regression

Following the resolution of multicollinearity issues, backward stepwise regression was applied to remove statistically insignificant predictors. At each step, the predictor with the highest p-value was eliminated, after which the model was refitted and re-evaluated.

This process required three iterations:

- **Removal 1:** `hepB`
- **Removal 2:** `population`
- **Removal 3:** `measles`

After these removals, the model was statistically significant(Table 4.3) and all the remaining predictors were also statistically significant at 5% level of significance(Table 4.2), and further elimination was not warranted.

4.1.4 Final Model

After resolving multicollinearity and applying backward stepwise regression, the final multiple linear regression model retained seven statistically significant predictors: `schooling`, `gdp`, `adMortality`, `HIV`, `thin9`, `polio`, and `status`. The fitted regression model is given in Equation 4.2.

$$\begin{aligned} \hat{le} = \hat{\beta}_0 &+ \hat{\beta}_1\,\texttt{schooling} + \hat{\beta}_2\,\log(\texttt{gdp}) + \hat{\beta}_3\,\texttt{adMortality} \\ &+ \hat{\beta}_4\,\log(\texttt{HIV}) + \hat{\beta}_5\,\log(\texttt{thin9}) \\ &+ \hat{\beta}_6\,\log(\texttt{polio}) + \hat{\beta}_7\,\text{status}. \end{aligned} \tag{4.2}$$

Table 4.2 presents the estimated regression coefficients, their standard errors, associated test statistics, and significance levels. Table 4.3 summarizes the overall goodness-of-fit measures for the final model and Table 4.4 shows the VIF's for the final model.

Table 4.2: Regression Coefficient Estimates for the Final Model

Predictor	**Estimate**	**Std. Error**	**t value**	**Pr($>\|t\|$)**
Intercept	61.80894	2.48763	24.847	$< 2 \times 10^{-16}$
schooling	0.70106	0.12115	5.787	3.24×10^{-08}
log(gdp)	0.34388	0.17387	1.978	0.0495
adMortality	-0.25619	0.03173	-8.073	1.06×10^{-13}
log(HIV)	-1.66975	0.27035	-6.176	4.46×10^{-09}
log(thin9)	-1.18255	0.26420	-4.476	1.37×10^{-05}
log(polio)	-0.94106	0.19836	-4.744	4.33×10^{-06}
statusDeveloping	-1.54782	0.70869	-2.184	0.0303

Table 4.3: Model Summary Statistics

Residual Standard Error	2.912 (df = 175)
Multiple R^2	0.8764
Adjusted R^2	0.8715
F-statistic	177.3 (df = 7, 175)
p-value	$< 2.2 \times 10^{-16}$

Table 4.4: VIF table for the final model

`schooling`	`gdp`	`adMortality`	`HIV`	`thin9`	`polio`	`status`
2.52	1.40	2.05	2.46	1.65	1.52	1.56

So, we will predict the Life Expectancy using the Equation 4.3

$$\begin{aligned} \hat{le} = 61.80 &+ 0.701\,\texttt{schooling} + 0.344\,\log(\texttt{gdp}) - 0.256\,\texttt{adMortality} \\ &- 1.670\,\log(\texttt{HIV}) - 1.183\,\log(\texttt{thin9}) \\ &- 0.941\,\log(\texttt{polio}) - 1.55\,\text{status}. \end{aligned} \tag{4.3}$$

4.2 Diagnostic Checking

To assess the validity of the linear regression assumptions, we examined the standard diagnostic plots: Residuals vs. Fitted, Q–Q plot, Scale–Location plot, and Residuals vs. Leverage. Overall, the results suggest that the model is reasonably appropriate(Figure diagnostic1).

The Residuals vs. Fitted plot shows a generally random scatter around zero, indicating no strong evidence of nonlinearity, although a slight curvature suggests that some nonlinear effects

may still be present but It's not so evident. The Q–Q plot shows that most standardized residuals fall close to the Normal line, implying that the normality assumption is largely satisfied, with only a few observations (e.g., point 183) deviating in the upper tail.

The Scale–Location plot demonstrates relatively constant variance across fitted values, though there is mild dispersion for mid-range fitted values, indicating weak heteroscedasticity but that is not severe. Finally, the Residuals vs. Leverage plot identifies a few high-leverage or influential points (notably observations 183 and 87), but none appear to exceed Cook's distance threshold, suggesting that no single observation unduly drives the model fit.

Overall, the diagnostics support the use of the linear regression model, with only minor concerns such as slight heteroscedasticity and a few moderately influential data points. So, it can be said that the Final model(4.2) mostly satisfy the the assumptions regarding linear regression.

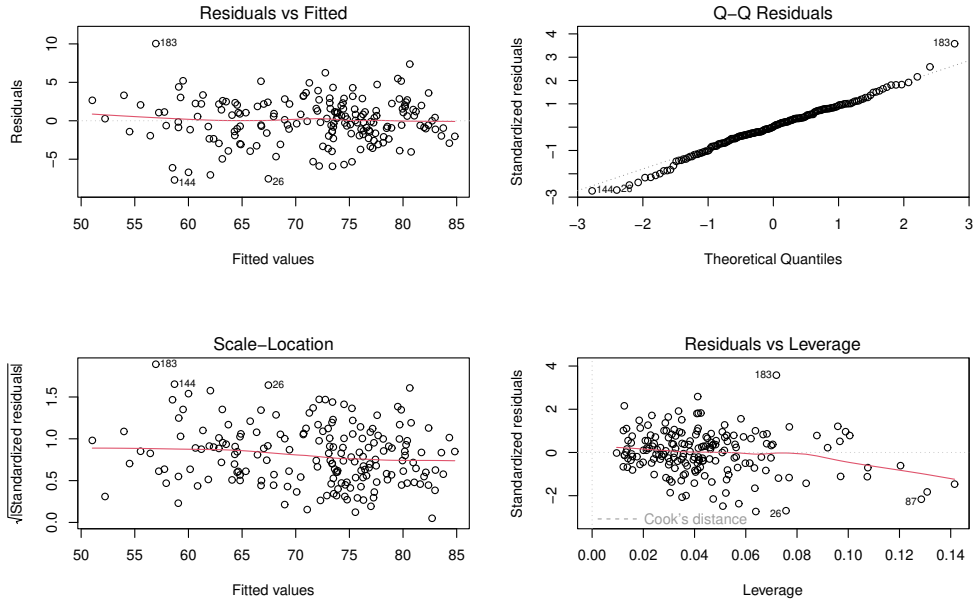


Figure 4.1: Model Diagnostic

Chapter 5

Results

This chapter presents and interprets the key findings of the final Multiple Linear Regression (MLR) model developed in Chapter 4. The final model successfully reduced multicollinearity and retained seven statistically significant predictors, providing a robust explanation for global variations in Life Expectancy.

5.1 Overall Model Performance

The summary statistics, detailed in Table 5.1 (corresponding to Table 4.3 in the previous chapter), demonstrate that the final model is highly significant and exhibits strong predictive power.

Table 5.1: Summary Statistics of the Final Model

Metric	**Value**
Residual Standard Error	2.912 ($df = 175$)
Multiple R^2	0.8764
Adjusted R^2	0.8715
F-statistic	177.3 ($df = 7, 175$)
p-value	$< 2.2 \times 10^{-16}$

5.1.1 Significance and Explanatory Power

The F-statistic of 177.3 with an extremely low p-value ($< 2.2 \times 10^{-16}$) is highly significant. This indicates that the final set of predictors collectively provides a substantially better fit to the data than a model with no predictors, confirming the overall statistical validity of the model.

The Adjusted R^2 value of **0.8715** suggests that the seven retained predictors (Average years of Schooling, GDP per capita, Adult Mortality Rate, HIV Death Rate, Early Childhood Thinness, Polio Immunization Deficit, and Economic Status) account for approximately **87.15**% of the total variance observed in Life Expectancy across the 183 countries. This high value confirms the model's excellent goodness-of-fit and predictive strength.

5.2 Interpretation of Regression Coefficients

The estimated coefficients for the final model are presented in Table 5.2 (corresponding to Table 4.2 in the previous chapter). All seven variables were retained due to their statistical significance at the 5% level, meaning each factor has a unique, independent contribution to Life Expectancy after controlling for all other factors. The final fitted model equation is:

$$\begin{aligned}\widehat{\text{Life Expectancy}} = 61.80 &+ 0.701\,\text{Schooling years} + 0.344\,\log(\text{GDP per capita}) \\ &- 0.256\,\text{Adult Mortality Rate} - 1.670\,\log(\text{HIV Death Rate}) \\ &- 1.183\,\log(\text{Early Childhood Thinness}) \\ &- 0.941\,\log(\text{Polio Immunization Deficite}) \\ &- 1.548\ \text{Economic Status(Developing)}\end{aligned} \tag{5.1}$$

Table 5.2: Regression Coefficient Estimates for the Final Model

Predictor	**Estimate**	**Std. Error**	t **value**	$Pr(>\|t\|)$
Intercept	61.80894	2.48763	24.847	$< 2 \times 10^{-16}$
schooling	0.70106	0.12115	5.787	3.24×10^{-08}
log(gdp)	0.34388	0.17387	1.978	0.0495
adMortality	−0.25619	0.03173	−8.073	1.06×10^{-13}
log(HIV)	−1.66975	0.27035	−6.176	4.46×10^{-09}
log(thin9)	−1.18255	0.26420	−4.476	1.37×10^{-05}
log(polio)	−0.94106	0.19836	−4.744	4.33×10^{-06}
status Developing	−1.54782	0.70869	−2.184	0.0303

5.2.1 Socioeconomic and Demographic Predictors

Years of Schooling (`schooling`)

The coefficient for `schooling` is $\hat{\beta}_{\text{schooling}} = \mathbf{0.701}$. This indicates a strong positive relationship: for every one-year increase in the average years of schooling completed by the adult population, Life Expectancy is expected to increase by approximately **0.701** years = **259.15** days, assuming all other health and economic variables remain constant. This highlights education's profound independent role as a socioeconomic determinant of longevity.

GDP per Capita (log(gdp))

The log-transformed GDP per capita has a positive coefficient of $\hat{\beta}_{\log(\text{gdp})} = \mathbf{0.344}$. For a log-transformed predictor, the interpretation is relative (More details in Subsection 2.1.2): an 1% *relative* increase in a country's raw GDP per capita is associated with an expected increase in Life Expectancy of approximately $(\frac{1}{100}) \times 0.344 = \mathbf{0.00344}$ years = **1.25** days, holding all other factors constant.

Country Status (`status Developing`)

The categorical variable `status` is a binary indicator (where 1 = Developing Country, 0 = Developed Country). The negative coefficient, $\hat{\beta}_{\text{Developing}} = \mathbf{-1.548}$, means that, on average,

a Developing Country is expected to have a Life Expectancy that is **1.548** years lower than a Developed Country, even after controlling for differences in schooling, GDP, and the specific health burdens measured in the model.

5.2.2 Health-Related Predictors

All health burden indicators retained in the model exhibit the expected statistically significant negative association with Life Expectancy.

Adult Mortality Rate (`adMortality`)

The Adult Mortality Rate coefficient is $\hat{\beta}_{\text{adMortality}} = \mathbf{-0.256}$. This is the strongest linear negative predictor: a 1% increase (the original adult mortality was in %) in the Adult Mortality Rate is expected to decrease Life Expectancy by **0.256** years = **93.4** days on average while other variables remain constant.

HIV Death Rate ($\log(\mathbf{HIV})$)

The log-transformed HIV Death Rate has a coefficient of $\hat{\beta}_{\log(\text{HIV})} = \mathbf{-1.670}$. This coefficient highlights the severe negative impact of the HIV epidemic on longevity. Approximately, a 1% *relative* increase in the raw HIV Death Rate is associated with a decrease in Life Expectancy of $(\frac{1}{100}) \times 1.670 = \mathbf{0.0167}$ years = **6.10** days on average while other factors remain constant.

Early Childhood Thinness ($\log(\mathbf{thin9})$)

The coefficient for log-transformed Early Childhood Thinness is $\hat{\beta}_{\log(\text{thin9})} = \mathbf{-1.183}$. This shows that chronic malnutrition in children aged 5–9 has a significant adverse effect. A 1% *relative* increase in the prevalence of Early Childhood Thinness is associated with a decrease in Life Expectancy of approximately $(\frac{1}{100}) \times 1.183 = \mathbf{0.01183}$ years = **4.32** days on average while others factors remain constant.

Polio Immunization Deficit ($\log(\mathbf{polio})$)

The log-transformed Polio Immunization Deficit has a coefficient of $\hat{\beta}_{\log(\text{polio})} = \mathbf{-0.941}$. This coefficient indicates that gaps in routine immunization coverage significantly predict lower longevity. A 1% increase in the percentage of children not immunized against Polio is associated with a decrease in Life Expectancy of approximately $(\frac{1}{100}) \times 0.941 = \mathbf{0.00941}$ years = **3.43** days on average while other variables remain constant, suggesting that basic preventative health measures are vital components of national longevity.

Chapter 6

Conclusion and Recommendations

6.1 Conclusion

The objective of this study was to quantify the independent influence of a comprehensive set of health and socioeconomic factors on Life Expectancy across 183 countries in 2015. Through the application of Multiple Linear Regression (MLR) and a rigorous model selection process involving multicollinearity checks and backward stepwise elimination, a final parsimonious model was developed.

The final model, presented in Chapter 5, achieved an Adjusted R^2 of **0.8715**, indicating that 87.15% of the variability in Life Expectancy is explained by the seven retained predictors: Adult Mortality Rate, Average Years of Schooling, HIV Death Rate, Early Childhood Thinness, Polio Immunization Deficit, GDP per capita, and Economic Status. This extremely high explanatory power confirms the model's robustness and the profound connection between these core indicators and national longevity.

The most critical findings underscore that health burdens, particularly the Adult Mortality Rate, and chronic disease, specifically the HIV Death Rate, exert the largest and most immediate negative pressures on Life Expectancy. Conversely, Average Years of Schooling stands out as the most powerful positive socioeconomic predictor in the model. The model also quantified the global disparity, showing that simply being classified as a Developing Country is associated with a 1.548 years reduction in Life Expectancy, even after controlling for specific socioeconomic and health differences.

In summary, this research confirms that improving global longevity requires coordinated investments across three primary domains: strengthening public health infrastructure, expanding educational attainment, and mitigating the effects of poverty and malnutrition, particularly those impacting child and adolescent health.

6.2 Recommendations

Based on the highly significant coefficients and the clear independent effects demonstrated by the final regression model, the following evidence-based policy recommendations are proposed to enhance Life Expectancy:

6.2.1 Recommendations for Public Health Infrastructure

- **Target Adult Mortality Reduction:** Given that the Adult Mortality Rate has the strongest linear negative correlation with Life Expectancy, policies must focus on improving access to high-quality primary care, chronic disease management (e.g., hypertension, diabetes), and injury prevention programs. Decreasing premature adult deaths offers the highest return on investment in longevity.
- **Prioritize HIV/AIDS Treatment and Prevention:** The large negative coefficient for the HIV Death Rate necessitates continued aggressive funding for anti-retro-viral treatment (ART) programs, expanding diagnostic testing accessibility, and intensifying mother-to-child transmission prevention efforts.
- **Ensure Universal Immunization Coverage:** The significant negative effect of the Polio Immunization Deficit highlights that failures in routine vaccination campaigns erode overall health. Governments should invest in logistics, cold chain management, and community outreach to achieve and maintain 100% coverage for core vaccines (e.g., Polio, Diphtheria, Hepatitis B).
- **Address Malnutrition and Early Childhood Thinness:** Chronic malnutrition, proxied by Early Childhood Thinness, significantly depresses Life Expectancy. Policy interventions should include national nutritional programs, mandatory nutrient fortification of staple foods, and targeted feeding programs for vulnerable age groups (5-9 years) to prevent long-term cognitive and physical deficits.

6.2.2 Recommendations for Systemic and Economic Factors

- **Expand Educational Access and Retention:** The large positive coefficient for Average Years of Schooling strongly recommends that educational policies move beyond mere enrollment to focus on maximizing the average years of schooling completed by the adult population. This includes reducing dropout rates, promoting adult literacy, and subsidizing secondary and vocational training, recognizing education's long-term and potent effect on health literacy and economic stability.
- **Leverage Economic Growth for Public Good:** While log-transformed GDP showed a diminishing return on Life Expectancy, national wealth remains a significant factor. Governments should ensure that GDP growth translates directly into public service improvements, particularly in health, sanitation, and clean water access, thereby minimizing the impact of the "Developing Country" status disparity.

6.2.3 Limitations and Scope for Future Work

The current model provides a robust and highly explanatory snapshot of Life Expectancy determinants in 2015. However, acknowledging its limitations is essential for guiding future research:

- **Cross-Sectional Data Limitation:** The analysis is based on a single year (2015). Future work should incorporate time-series data or panel data analysis to model the longitudinal impact of policy changes and economic fluctuations on Life Expectancy, establishing causality rather than mere correlation.

- **Missing Non-Observable Factors:** The model, despite its high R^2, does not account for critical non-observable determinants. Future research could enhance predictability by including variables related to:
    1. **Genetic and Environmental Factors:** The current health indicators focus on infectious disease and mortality rates. Integrating data on common genetic diseases, air and water quality, and climate change effects could significantly increase the model's predictive accuracy.
    2. **Healthcare System Quality:** The model lacks detailed metrics on the quality and efficiency of healthcare delivery (e.g., emergency response times, doctor-to-patient ratios, medical technology accessibility), which are likely strong determinants of longevity.
- **Alternative Modeling Techniques:** While MLR provided a clear and interpretable result, exploring non-linear and machine learning approaches, such as Generalized Additive Models (GAM) or Random Forests, could capture complex, non-additive interactions between predictors that the current linear framework may miss.

THE END